

applications (as given below), it will help users understand and appreciate the beauty of the former. In microservices:

- One single application is broken down into a network of processes.
- All these services communicate using REST or MQ.
- Applications are loosely coupled.
- The scaling up of applications is a lot easier.
- There is very good isolation of these services, as when one fails, others can continue.

What is Kubernetes and why should one use it?

Kubernetes is an open source orchestrator for deploying containerised applications (microservices). It is also defined as a platform for creating, deploying and managing various distributed applications. These applications may be of different sizes and shapes. Kubernetes was originally developed by Google to deploy scalable, reliable systems in containers via application-oriented APIs. Kubernetes is suitable not only for Internet-scale companies but also for cloud-native companies, of all sizes. Some of the advantages of Kubernetes are listed below:

- Kubernetes provides the software necessary to build and deploy reliable and scalable distributed systems.
- Kubernetes supports container APIs with the following benefits:
 - Velocity—a number of things can be shipped quickly, while also staying available.
 - Scaling—favours scaling with decoupled architecture through load balancers and scaling with consistency.
 - Abstract—applications built and deployed on top of Kubernetes can be ported across different environments. Developers are separated from specified machines for providing abstraction. This reduces the overall machines required, thus reducing the cost of CPUs and RAM.
 - Efficiency—the developer's test environment can be cheaply and quickly created via Kubernetes clusters and this can be shared as well, thus reducing the cost of development.

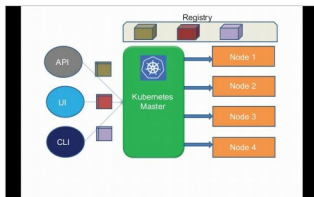


Figure 2: Kubernetes architecture

- Kubernetes continuously takes action to ensure that the current state matches the desired state.

The various components involved in Kubernetes

Pods: These are groups of containers that can group together other container images developed by different teams into a single deployable unit.

NameSpaces: This provides isolation and complete access control for each microservice, to control the degree to which other services interact with it.

Kubernetes services: Provides load balancing, discovery isolation and naming of microservices.

Ingress: These are objects that provide an easy-to-use front-end (externalised API surface area).

Running and managing containers using Kubernetes

As described earlier, Kubernetes is a platform for creating, deploying and managing distributed applications. Most of these applications take an input, process the data and provide the results as output. Most of these applications contain language runtime, libraries (*libc* and *libssl*) and source code.

A container image is a binary package that encapsulates all of the files necessary to run an application inside an OS container. The Open Container Image (OCI) is the standard image format that's most widely used.

The container types are of two categories:

- System containers, which try to imitate virtual machines and may run the full boot processes
 - Application containers, which run single applications
- These images can be run using the `docker run -d --name` command, using the CLI.

The default container runtime used by Kubernetes is Docker, as the latter provides an API for creating application containers on both Linux and Windows based operating systems.

kuard is a database and maps to Port 8080, and can also be explored using a Web interface.

Docker provides many features by exposing the underlying 'cgroups' technology provided by the Linux kernel. With this, the following resource usage can be managed and monitored:

- Memory resources management and limitation
- CPU resources management and limitation

Deploying a Kubernetes cluster

Kubernetes can be installed on three major cloud providers like that of Amazon's AWS, Microsoft's Azure and Google's Cloud Platform (GCP). Each cloud provider allows its own container service platforms.

Kubernetes also can be installed using Minikube, locally. Minikube is a simulation of the Kubernetes cluster, but the main function of this is for experimentation, local